SIMATS Engineering

SIMATS Online Programming Lab — Python Programming

Course Name	Python Programming
Course Code	CSA0804
Name	Monish.L
Register Number	1972260035B
Topic	CO5-AT2-DEBUGGING AND OPTIMIZATION TASK
Question	Q1 — Buggy CSV Data Loader (5 Marks) — Pandas – CSV, Data Types

Q1. Buggy CSV Data Loader [5 Marks] — Pandas – CSV, Data Types

■ This program contains 4 bug(s). Identify, fix, and optimise.

(a) Identify ALL errors in the program below.

(b) Rewrite the corrected and optimised version.

(c) Add code to display: shape, dtypes, null count, and basic describe() statistics.

Buggy Program:

```
import panda as pd
df = pd.read_csv('sales_data')

df['Revenue'] = df['Revenue'].astype('string')
df.dropna(inplace = False)
print(df.head(10))
print(df.describe)
```

(a) Errors Identified

Bug 1 – Import statement (Line 1):

import panda as pd is incorrect. The library is named **pandas**, not **panda**. This raises a **ModuleNotFoundError** and stops the script before it can run.

Bug 2 – Missing file extension (Line 2):

pd.read_csv('sales_data') is missing the **.csv** extension. Pandas will look for a file literally named *sales_data* and raise a **FileNotFoundError**. It should be **'sales_data.csv'**.

Bug 3 – Wrong data type conversion (Line 4):

df['Revenue'] = df['Revenue'].astype('string') converts a numeric column into text. Revenue must stay numeric to be summed, plotted, or summarised with **describe()**; as a string column it would be excluded from numeric statistics entirely. It should be converted with **pd.to_numeric()** instead.

Bug 4 – dropna() result discarded (Line 5):

df.dropna(inplace = False) returns a new, cleaned DataFrame but the result is never captured, and **inplace = False** means the original **df** is left unchanged. Nulls silently remain in the data. It must be reassigned, e.g. **df = df.dropna()**.

Note: **print(df.describe)** on the final line is also missing its parentheses (**df.describe()**) — without them it prints the method object instead of calling it. This is fixed as part of part (c) below, alongside the new statistics code that was added.

(b) Corrected and Optimised Program

The program below fixes all four bugs, converts Revenue to a proper numeric type, safely drops missing values, and adds the diagnostic output requested in part (c).

```
"""
CO5-AT2 - Debugging and Optimisation Task
Challenge 1: Buggy CSV Data Loader [Pandas - CSV, Data Types]
------------------------------------------------------------
Loads sales data from CSV, converts the Revenue column to a
numeric type, removes rows with missing values, and prints
diagnostic information: shape, dtypes, null counts, and
summary statistics.
"""

import pandas as pd

# ---------------------------------------------------------------
# 1. Load the CSV file (fixed: correct import + file extension)
# ---------------------------------------------------------------
df = pd.read_csv('sales_data.csv')

# ---------------------------------------------------------------
# 2. Fix data types (fixed: Revenue must stay numeric, not text)
# ---------------------------------------------------------------
df['Revenue'] = pd.to_numeric(df['Revenue'], errors='coerce')

# ---------------------------------------------------------------
# 3. Drop missing values (fixed: reassign the cleaned DataFrame)
# ---------------------------------------------------------------
df = df.dropna()

# ---------------------------------------------------------------
# 4. Preview the cleaned data
# ---------------------------------------------------------------
print("First 10 rows:")
print(df.head(10))

# ---------------------------------------------------------------
# 5. Diagnostics required by part (c): shape, dtypes, null
#    count, and describe() statistics
# ---------------------------------------------------------------
print("\nShape (rows, columns):", df.shape)

print("\nColumn data types:")
print(df.dtypes)

print("\nNull value count per column:")
print(df.isnull().sum())

print("\nSummary statistics:")
```

```
print(df.describe())
```

(c) Diagnostic Code Added

- **df.shape** – prints the (rows, columns) tuple so the dataset size is visible at a glance.
- **df.dtypes** – prints the data type of every column, confirming Revenue is now numeric.
- **df.isnull().sum()** – prints the count of missing values remaining in each column (0 after dropna()).
- **df.describe()** – called with parentheses, prints count, mean, std, min, quartiles, and max for the numeric columns.

Expected Sample Output

Exact values depend on the contents of **sales_data.csv**. For a file with columns such as *Date, Product, Revenue* and 12 rows (2 containing missing values), the output would look like:

```
First 10 rows:
         Date   Product  Revenue
0  2026-01-01    Widget    120.0
1  2026-01-02    Gadget    250.0
2  2026-01-03    Widget    180.0
...

Shape (rows, columns): (10, 3)

Column data types:
Date        object
Product     object
Revenue    float64
dtype: object

Null value count per column:
Date       0
Product    0
Revenue    0
dtype: int64

Summary statistics:
          Revenue
count   10.000000
mean   195.400000
std     48.216000
min    120.000000
25%    162.500000
50%    195.000000
75%    227.500000
max    280.000000
```

Conclusion

The original loader failed before it could run due to a misspelled **pandas** import and a missing **.csv** extension, and even if those were overlooked it would have silently corrupted the Revenue column into text and left null values in place because the **dropna()** result was discarded. The corrected version fixes all four issues, keeps Revenue numeric with **pd.to_numeric()**, reassigns the cleaned DataFrame from **dropna()**, and adds the requested diagnostics (**shape**, **dtypes**, null counts, and **describe()**) so the dataset can be verified and summarised correctly.